

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – PSEUDOCODE WRITING

CO1A1

Course Code:	CSA0206
Course Title:	C Programming
CO Assessed:	CO1 – Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)
Assessment:	Assessment Tool 1 – Pseudocode Writing
Total Marks:	100

Student Name:	
Reg. No.:	
Section / Batch:	Date:

CODE OF CONDUCT

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

INSTRUCTIONS

- This assessment contains one question.
- Students should identify the required inputs, outputs, constraints, and develop a clear logical solution.
- Pseudocode should be well-structured, clear, consistent, and easy to follow.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. Students should provide written materials

RUBRICS FOR PSEUDOCODE WRITING

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

PSEUDOCODE QUESTIONS

A smart energy company wants to develop a C program to calculate the monthly electricity bill of a consumer. The program should begin with appropriate preprocessor directives and define a constant for the fixed service charge. The consumer ID, units consumed, and applicable tariff values must be stored using suitable variables and data types. The system should calculate the energy charge using arithmetic operators based on the units consumed. A fixed service charge must then be added to obtain the preliminary bill amount. If the preliminary bill exceeds a specified threshold, a surcharge percentage must be applied. The final bill should be calculated using an expression containing arithmetic, relational, and conditional operators. The programmer must ensure that operator precedence and associativity produce the intended calculation. The program should avoid unnecessary intermediate variables wherever a valid expression can be used. The pseudocode must clearly indicate the sequence of input, calculation, decision-making, and output operations. Finally, display the consumer ID, units consumed, energy charge, surcharge, and total payable amount. Write the pseudocode and corresponding C program, and justify the data types, operators, precedence, and associativity used in the solution.